Temperature, entropy, and thermal ionization

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2.1 What is temperature?

$$E \longrightarrow W(E) \longrightarrow S = k_B \ln W \longrightarrow \frac{1}{T} = \frac{\partial S}{\partial E}$$

energy — microstates — entropy — temperature

W multiplicity [1] k_B Boltzmann constant

2.1 Microstate counting

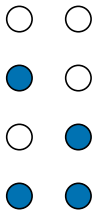
1 cell

$W = 2^1$ [1]



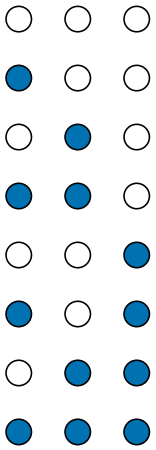
2 cells

$W = 2^2$ [1]



3 cells

$W = 2^3$ [1]



$W_1 = 2$

$W_2 = 4$

$W_3 = 8$

$$W_3 = W_2 W_1$$

4×2, not $4 + 2$

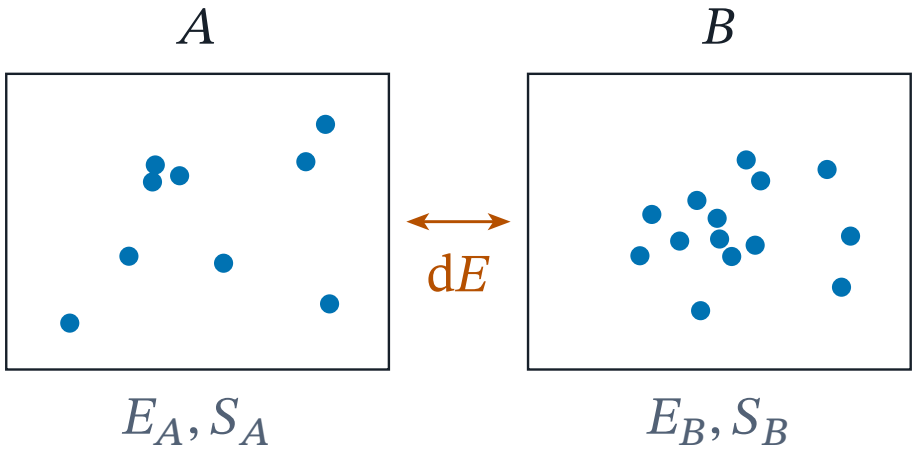
2.1 Additive entropy

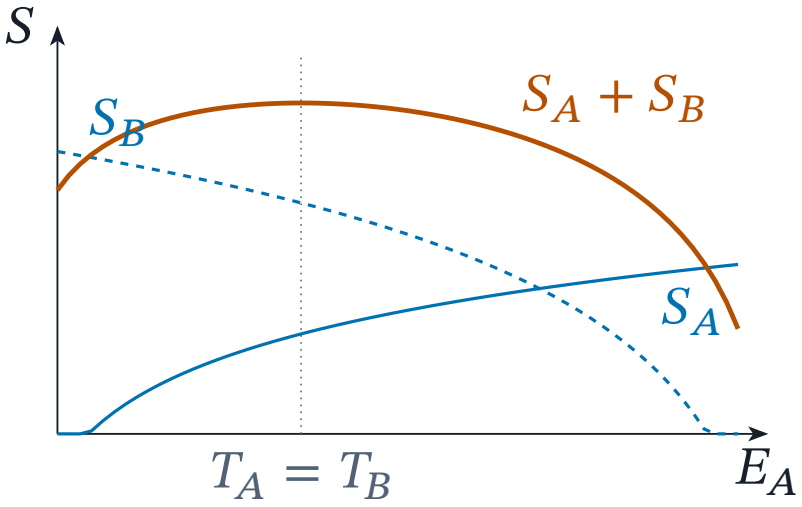
$$W_{A+B} = W_A W_B \quad \Longrightarrow \quad \ln W_{A+B} = \ln W_A + \ln W_B$$

$$S = k_B \ln W, \quad S_{A+B} = S_A + S_B$$

entropy, J/K

2.2 Two exchanging boxes





$$\mathrm{d}S_{\mathrm{tot}} = \left(\frac{1}{T_A} - \frac{1}{T_B}\right)\mathrm{d}E_A = 0 \quad \Longrightarrow \quad T_A = T_B$$

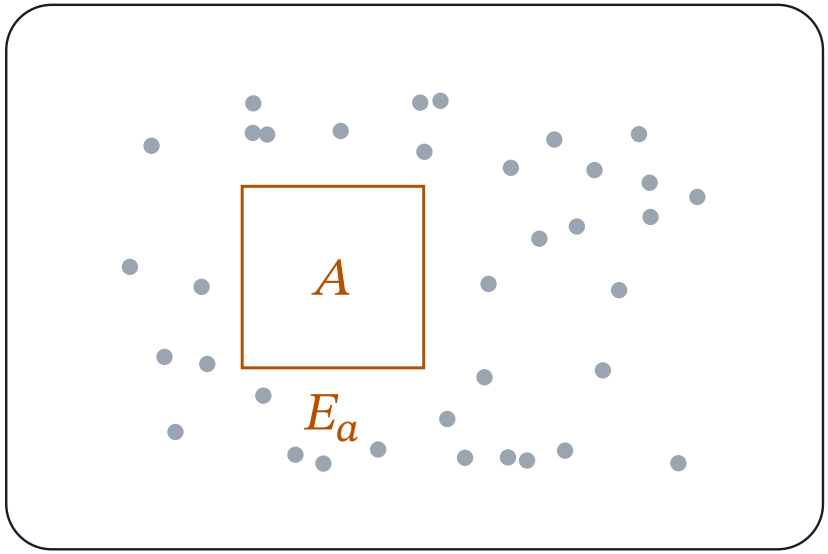
maximum entropy: equal temperatures

2.2 Energy balance

$$\mathrm{d}E = \underbrace{T\,\mathrm{d}S}_{\text{heat}} \underbrace{-p\,\mathrm{d}V}_{\text{work}} + \underbrace{\mu\,\mathrm{d}N}_{\text{particles}} + \underbrace{\boldsymbol{u}\cdot\mathrm{d}\boldsymbol{P}}_{\text{bulk flow}}$$

p pressure μ chemical potential $\boldsymbol{u}$ bulk velocity $\boldsymbol{P}$ total momentum $T > 0$

2.3 Reservoir weight



reservoir $E_{\text{tot}} - E_a$

$$P_a \propto W_{\text{rest}}(E_{\text{tot}} - E_a)$$

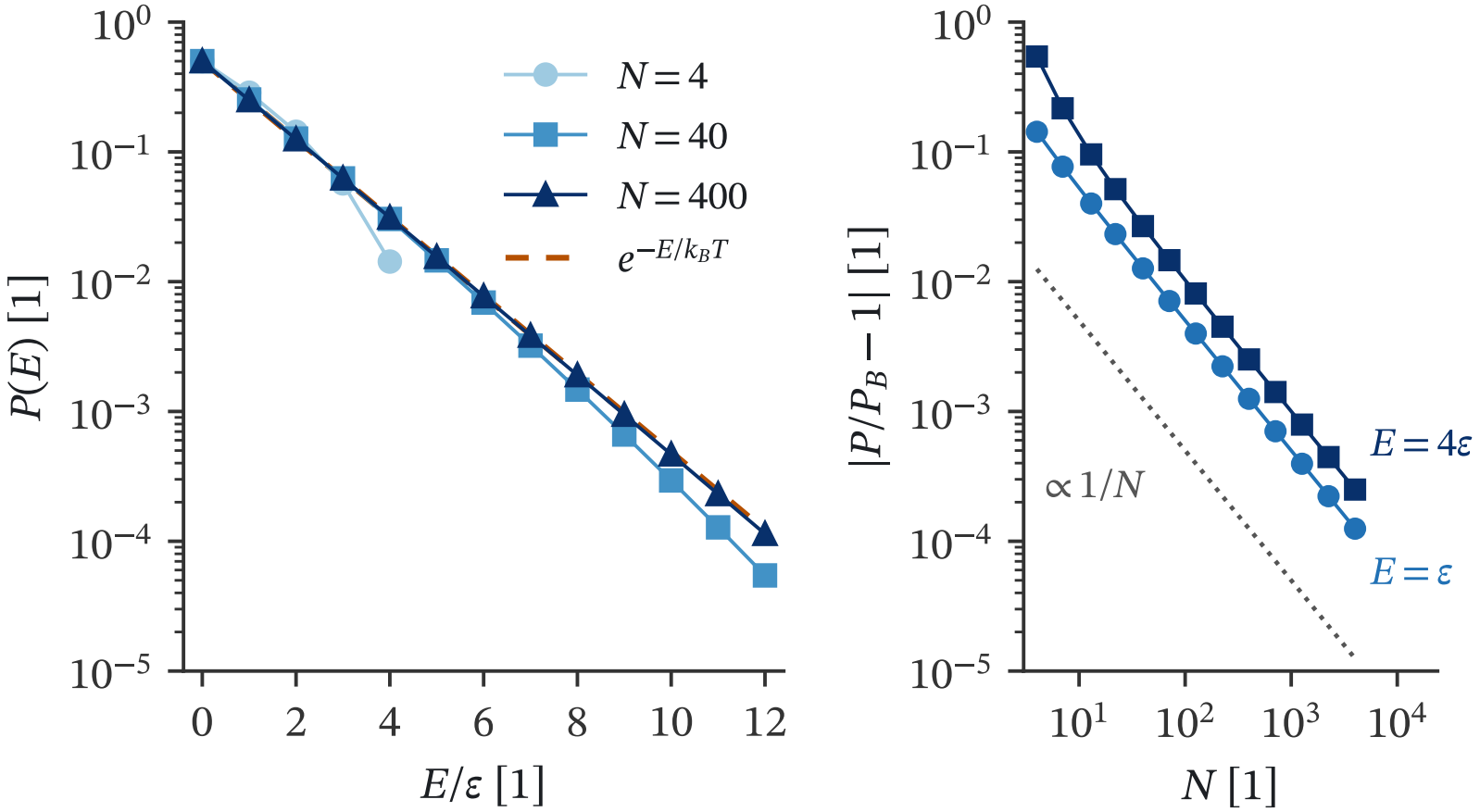
$$= \exp\left(\frac{S_{\text{rest}}(E_{\text{tot}} - E_a)}{k_B}\right)$$

$$S_{\text{rest}}(E_{\text{tot}} - E_a) \approx S_{\text{rest}}(E_{\text{tot}}) - \frac{E_a}{T}$$

$$P_a \propto \exp\left(-\frac{E_a}{k_B T}\right)$$

Boltzmann factor of microstate a

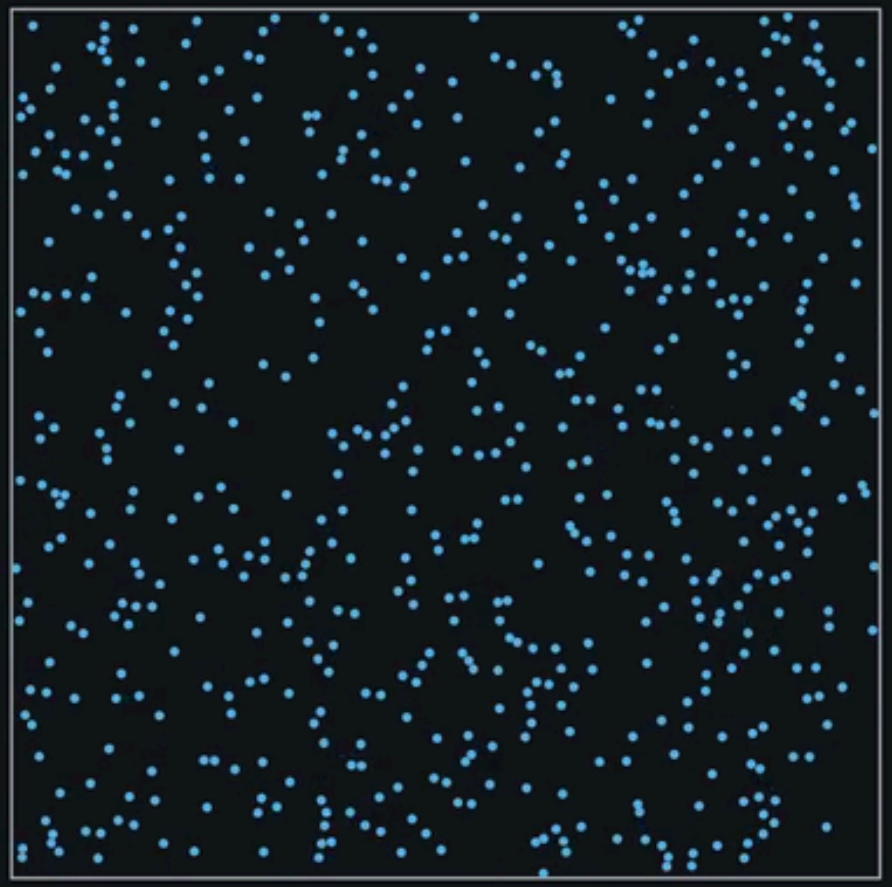
2.3 Boltzmann factor by counting



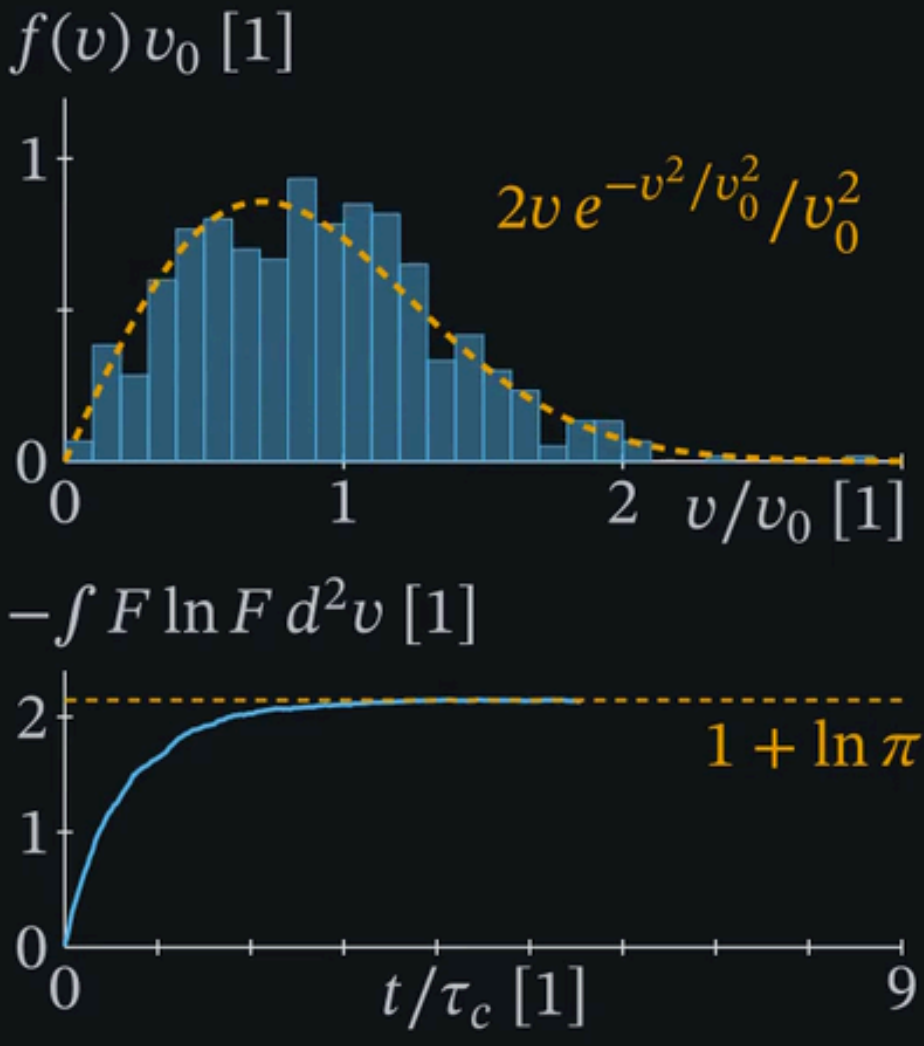
$$P(E) \propto W_{\text{rest}}(Q - E/\varepsilon), \quad W_{\text{rest}}(q) = \frac{(q+N-1)!}{q!\,(N-1)!}, \quad k_B T = \varepsilon / \ln 2$$

N reservoir oscillators sharing $Q = N$ quanta ε; P_B Boltzmann value

hard disks, $|\mathbf{v}| = v_0$ at $t = 0$



$|E/E_0 - 1| < 10^{-15}$ $t/\tau_c = 5.5$



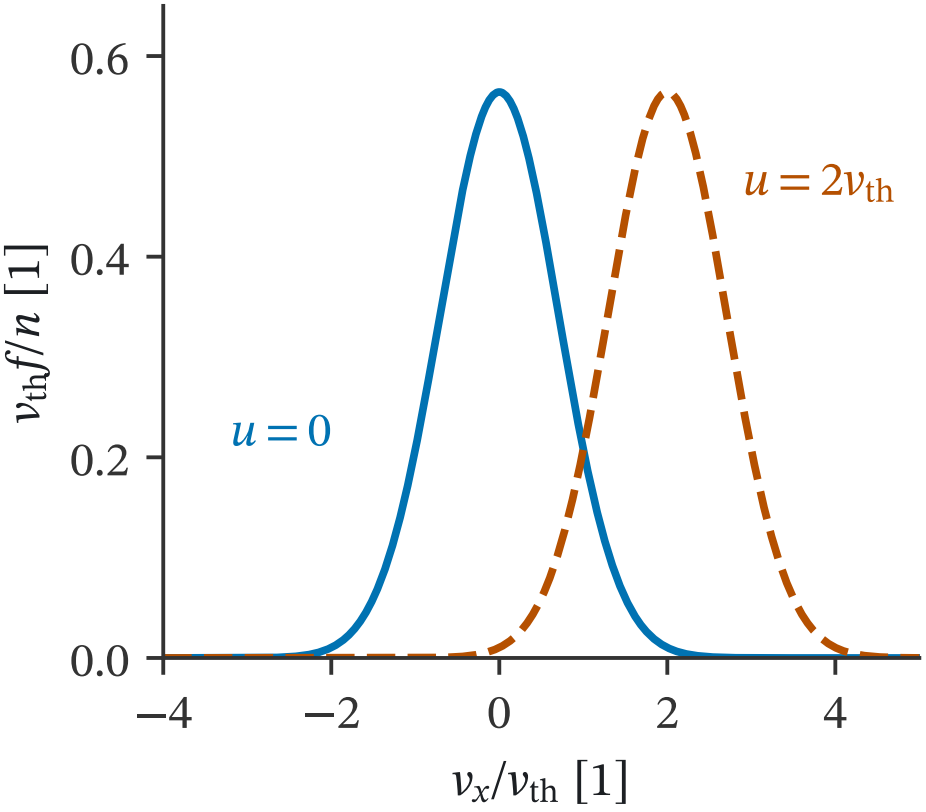
2.3 Drifting Maxwellian

$$\boldsymbol{u} = \frac{\boldsymbol{P}}{Nm}$$

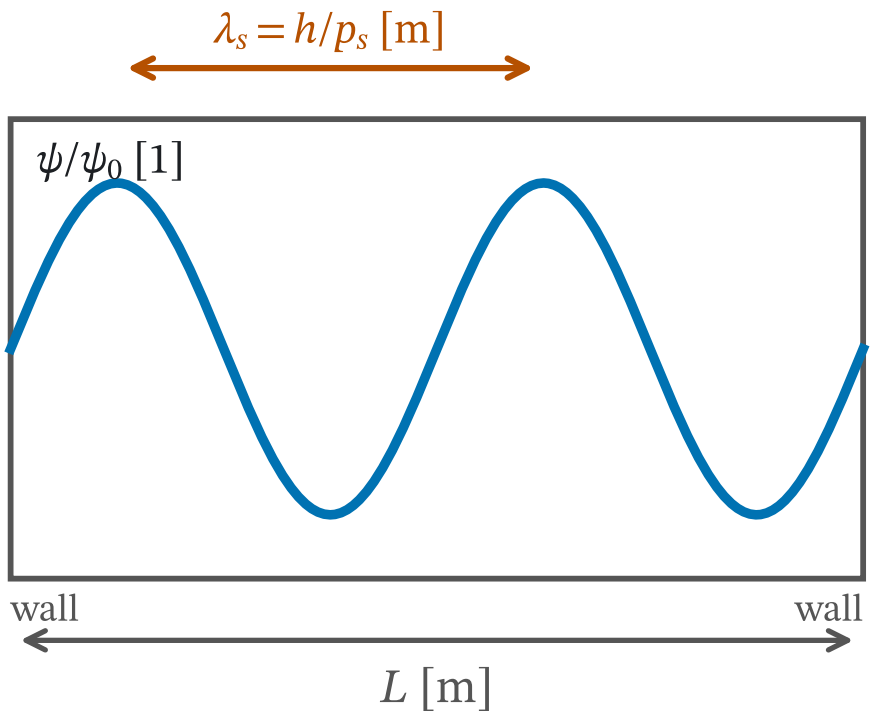
$$E_{\text{kin}} = \frac{\boldsymbol{P}^2}{2Nm} + E_{\text{random}}$$

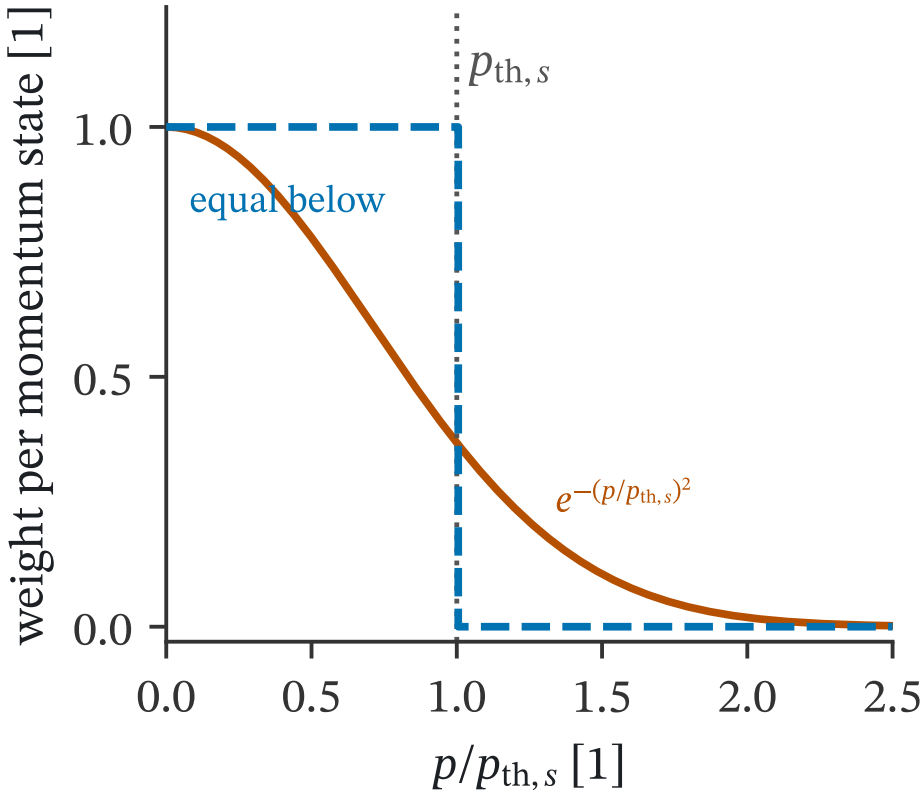
$$f_s \propto \exp\left(-\frac{(\boldsymbol{v} - \boldsymbol{u}_s)^2}{v_{\text{th},s}^2}\right)$$

$$v_{\text{th},s} = \sqrt{\frac{2k_B T_s}{m_s}}, \text{ §1.2}$$



2.4 Where the slots come from





$$M_s \approx \frac{g_s V}{\lambda_s^3}, \quad \lambda_s = \frac{h}{p_s}$$

one standing wave per de Broglie cell

$$\lambda_e \approx \frac{h}{\sqrt{2 m_e k_B T}} \longrightarrow \lambda_{\mathrm{th},e} = \frac{h}{\sqrt{2\pi m_e k_B T}}$$

sharp cutoff → Maxwellian weight: factor $\sqrt{\pi}$

2.4 Thermal particle slots

$$M_s = \frac{g_s V}{\lambda_{\mathrm{th},s}^3}$$

thermally available one-particle slots [1]

$$\frac{M_s}{N_s} = \frac{g_s}{n_s \lambda_{\mathrm{th},s}^3} \gg 1$$

dilute: many free slots per particle

$$\lambda_{\mathrm{th},s} = \frac{h}{\sqrt{2\pi m_s k_B T}}$$

$\lambda_{\mathrm{th},s}$ thermal de Broglie wavelength
g_s internal states at equal energy
h Planck constant

2.4 Adding and removing

$$W_s(N_s) \approx \frac{M_s^{N_s}}{N_s!}$$

add one particle

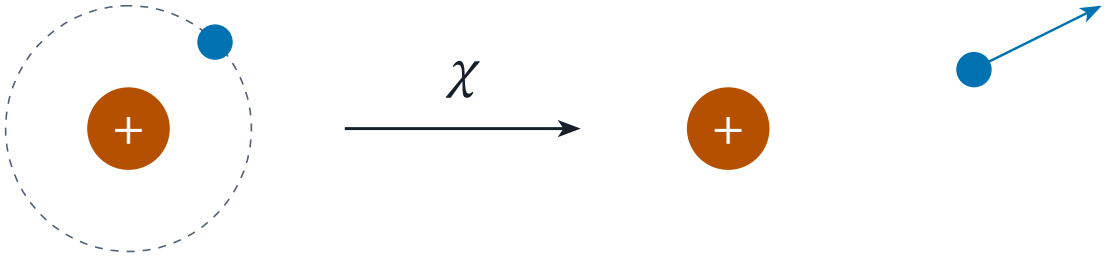
$$\frac{W_s(N_s+1)}{W_s(N_s)} = \frac{M_s}{N_s+1} \approx \frac{M_s}{N_s}$$

remove one particle

$$\frac{W_s(N_s-1)}{W_s(N_s)} = \frac{N_s}{M_s}$$

dilute and indistinguishable: $1 \ll N_s \ll M_s$

2.4 One ionization



neutral n ion i free electron e

$$\frac{n_n \lambda_{\mathrm{th},n}^3}{g_n} \qquad \frac{g_i}{n_i \lambda_{\mathrm{th},i}^3} \qquad \frac{2}{n_e \lambda_{\mathrm{th},e}^3} \qquad e^{-\chi/k_B T}$$

remove neutral | add ion | add electron | reservoir pays χ

$$R = \frac{W_{\text{after}}}{W_{\text{before}}} = \text{product of the four factors}, \quad R = 1 \text{ in equilibrium}$$

2.4 Heavy particle cancellation

$$m_n \approx m_i \implies \lambda_{\mathrm{th},n} \approx \lambda_{\mathrm{th},i}$$

$$R \approx \underbrace{\frac{n_n g_i}{n_i g_n}}_{\text{neutral} \to \text{ion}} \cdot \underbrace{\frac{2}{n_e \lambda_{\mathrm{th},e}^3}}_{\text{new free electron}} \cdot e^{-\chi/k_B T}$$

the genuinely new freedom is the free electron

2.4 Saha equation

$$\underbrace{\frac{n_i n_e}{n_n}}_{\text{composition}} = \underbrace{\frac{2}{\lambda_{\mathrm{th},e}^3}}_{\text{free-electron states}} \cdot \underbrace{\frac{g_i}{g_n}}_{\text{internal states}} \cdot \underbrace{e^{-\chi/k_B T}}_{\text{energy penalty}}$$

χ ionization energy $\quad g_i = g_n = 1$ here: order-one factors

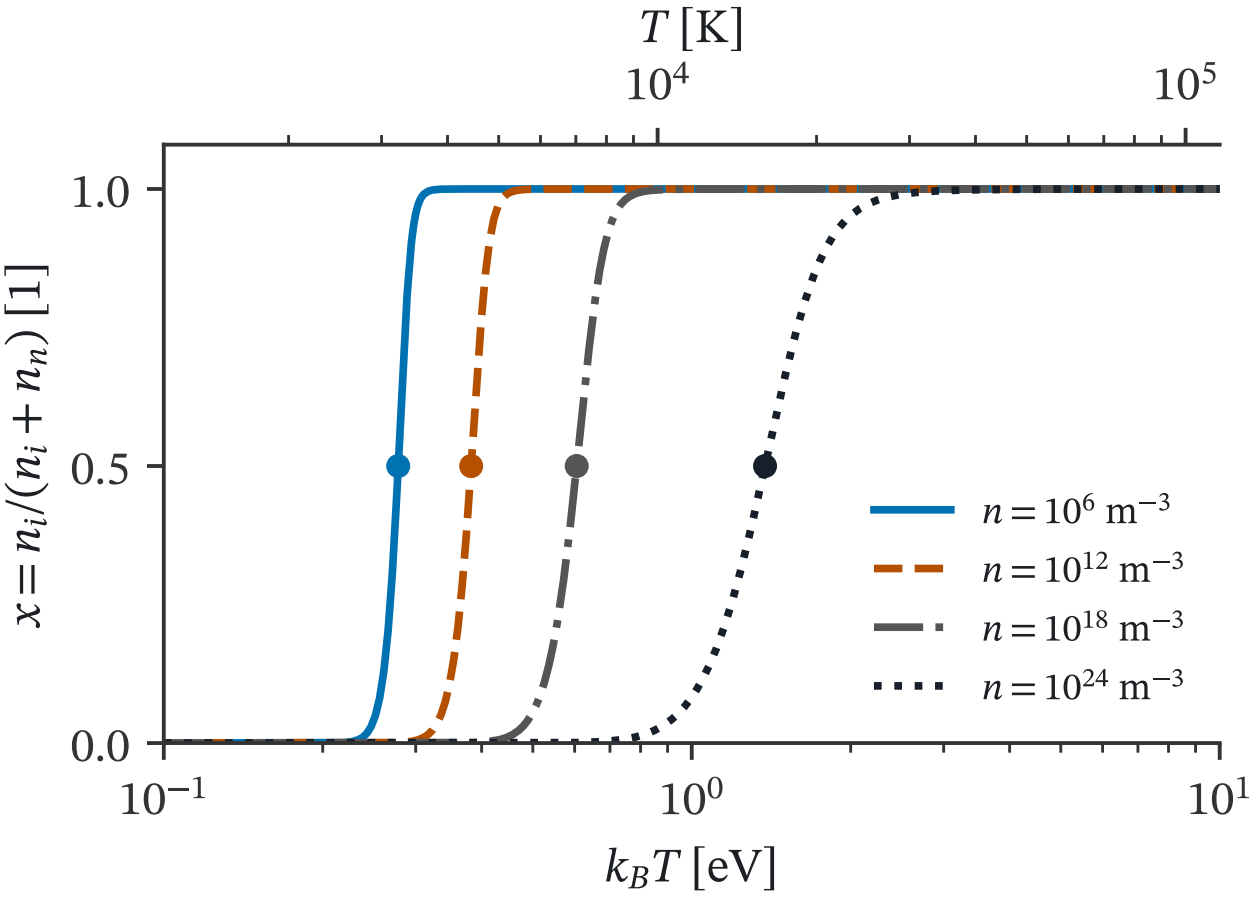
2.4 Saha equation in numbers

$$\frac{n_i n_e}{n_n} = 2\left(\frac{2\pi m_e k_B T}{h^2}\right)^{3/2} e^{-\chi/k_B T}$$

$$2\left(\frac{2\pi m_e k_B}{h^2}\right)^{3/2} = 4.83 \times 10^{21} \mathrm{m}^{-3}\mathrm{K}^{-3/2}$$

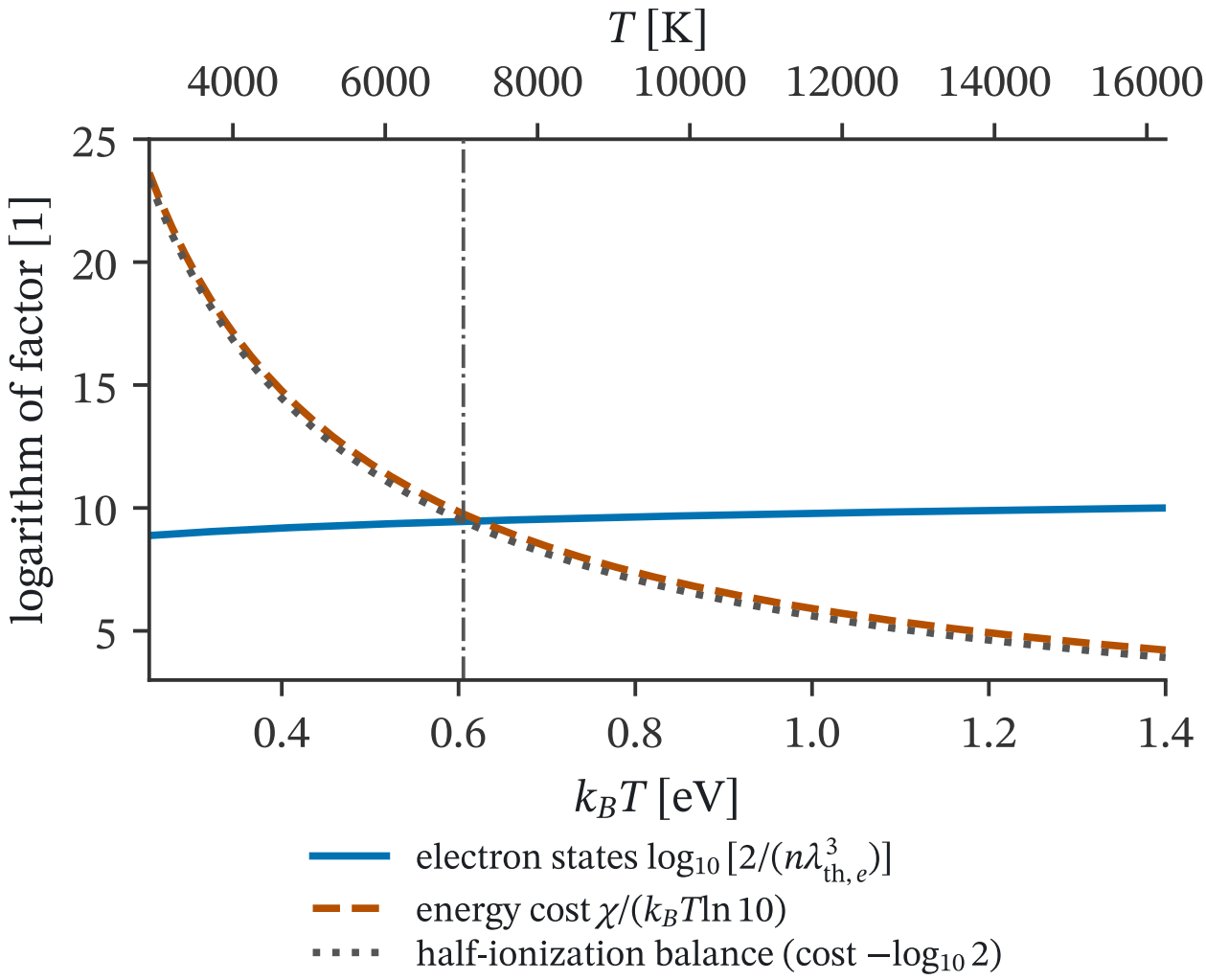
hydrogen: $\chi = 13.6$ eV, $\frac{\chi}{k_B} = 1.58 \times 10^5$ K

2.5 Ionization fraction



$$x = \frac{n_i}{n_i + n_n}, \quad n_e = n_i, \quad \frac{x^2}{1-x} = \frac{1}{n} \cdot \frac{2}{\lambda_{\mathrm{th},e}^3} e^{-\chi/k_B T}$$

2.5 Entropy against energy



half ionization where the free-electron states pay the energy penalty

2.5 Why about 10^4 K?

$$k_B T_{1/2} = \frac{\chi}{\ln\left(4/n\lambda_{\mathrm{th},e}^3\right)}$$

half ionization, $\chi = 13.6$ eV

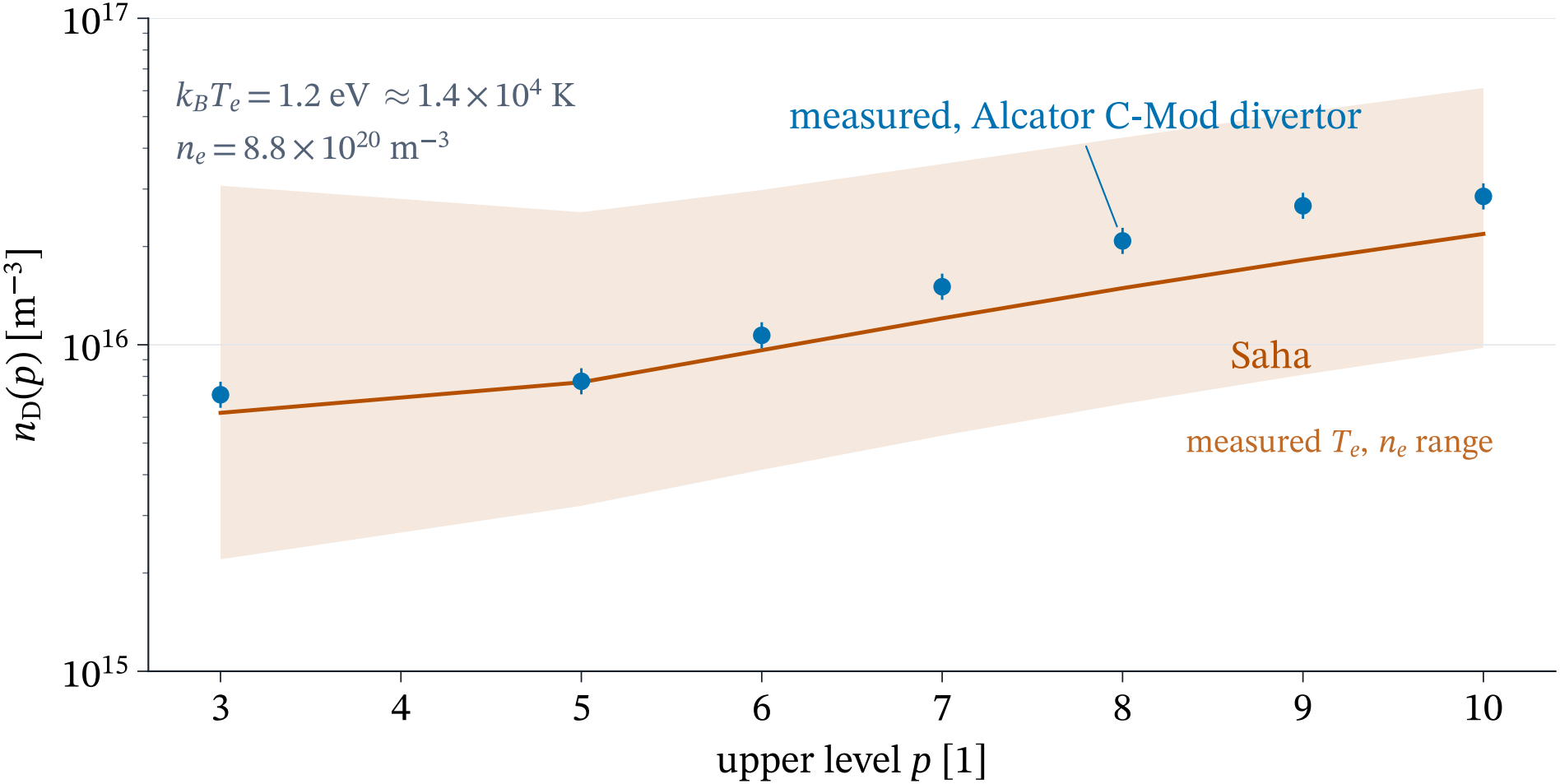
$\ln\left(4/n\lambda_{\mathrm{th},e}^3\right) \approx 10 \ldots 50$

$n = 10^{18} \ldots 10^{24}\,\mathrm{m}^{-3}: \quad T_{1/2} \approx$ 7000…16000 K

logarithm: weak dependence on density

n [m^{-3}]	$T_{1/2}$ [K]	$k_B T_{1/2}$ [eV]	$\chi/k_B T_{1/2}$
10^6	3200	0.28	49
10^{12}	4400	0.38	36
10^{18}	7000	0.61	22
10^{24}	16000	1.38	10
10^{26}	26200	2.26	6

2.5 Saha in a tokamak divertor



Excited deuterium levels in the detached Alcator C-Mod divertor, measured from Balmer lines, against Saha–Boltzmann at the measured $k_B T_e$ and Stark n_e (band: their uncertainty ranges). Data redrawn from D. Lumma, J. L. Terry, B. Lipschultz, Phys. Plasmas 4, 2555 (1997), doi:10.1063/1.872234; bars: digitization only.

2.5 Equilibrium validity

Saha
local thermodynamic equilibrium

dilute radiation escape
coronal / collisional-radiative balance

solar corona; tokamak edge
many laboratory discharges

nonthermal electrons; photons
rate-dependent ionization

high density
pressure ionization

ideal slots
$$n_s \frac{\lambda_{\mathrm{th},s}^3}{g_s} \ll 1$$

2.5 Quantum degenerate plasma

$$\frac{n_e \lambda_{\mathrm{th},e}^3}{2} \gtrsim 1$$

slots full: Pauli exclusion, Fermi statistics

plasma	n_e [m^{-3}]	T	$n_e \lambda_{\mathrm{th},e}^3/2$
tokamak core	10^{20}	10 keV	2×10^{-14}
compressed ICF fuel	2×10^{32}	200 eV	10
electrons in copper	8.5×10^{28}	300 K	3×10^3
white dwarf	10^{36}	10^7 K	7×10^3

$$\frac{n_e \lambda_{\mathrm{th},e}^3}{2} \ll 1 \quad \text{classical plasma}$$

same slot count as in the Saha equation

Credits

Physics: Reif (1965); Saha (1920), DOI 10.1080/14786441008636148.
Constants: NIST CODATA 2022.
Original diagrams and computed plots: Christopher Albert, CC BY 4.0.

AI tools assisted drafting and layout; scientific review by Christopher Albert.